

A-Level Maths:

Core 4

for Edexcel



C4.5 Integration 1



These icons indicate that teacher's notes or useful web addresses are available in the Notes Page.

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For more detailed instructions, see the *Getting Started* presentation.

- **Integrals of standard functions**
- Reversing the chain rule
- Integration by substitution
- Integration by parts
- Volumes of revolution
- Examination-style question



So far, we have only looked at functions that can be integrated using:

$$\int kx^n dx = \frac{kx^{n+1}}{n+1} + c \quad \text{for all } n \neq -1$$

For example:

Integrate $10x^4 + \frac{1}{x^2} + 6\sqrt{x} - 3$ with respect to x .

$$\begin{aligned} \int \left(10x^4 + \frac{1}{x^2} + 6\sqrt{x} - 3 \right) dx &= \int 10x^4 dx + \int x^{-2} dx + \int 6x^{\frac{1}{2}} dx - \int 3 dx \\ &= \frac{10x^5}{5} + \frac{x^{-1}}{-1} + \frac{6x^{\frac{3}{2}}}{\frac{3}{2}} - 3x + c \\ &= 2x^5 - \frac{1}{x} + 4\sqrt{x^3} - 3x + c \end{aligned}$$

The integral of $\frac{1}{x}$

The only function of the form x^n that cannot be integrated by this method is $x^{-1} = \frac{1}{x}$.

Adding 1 to the power and then dividing would lead to the meaningless expression,

$$\frac{x^0}{0}$$

This does not mean that $\frac{1}{x}$ cannot be integrated.

Remember that

$$\text{if } y = \ln x \text{ then } \frac{dy}{dx} = \frac{1}{x}$$

Therefore

$$\int \frac{1}{x} dx = \ln x + c \quad (\text{where } x > 0)$$



The integral of $\frac{1}{x}$

We can only find the log of a positive number and so this is only true for $x > 0$.

However, $\frac{1}{x}$ *does* exist for $x < 0$ (but not $x = 0$). So how do we integrate it for *all* possible values of x ?

We can get around this by taking x to be negative.

If $x < 0$ then $-x > 0$ so:

$$\int \frac{1}{x} dx = \int \frac{-1}{-x} dx = \ln(-x) + c \quad (\text{where } x < 0)$$

We can combine the integrals of $\frac{1}{x}$ for both $x > 0$ and $x < 0$ by using the modulus sign to give:

$$\int \frac{1}{x} dx = \ln|x| + c$$

The integral of $\frac{1}{x}$

Find $\int \frac{2}{3x} dx$

This is just the integral of $\frac{1}{x}$ multiplied by a constant.

$$\int \frac{2}{3x} dx = \frac{2}{3} \int \frac{1}{x} dx = \frac{2}{3} \ln|x| + c$$

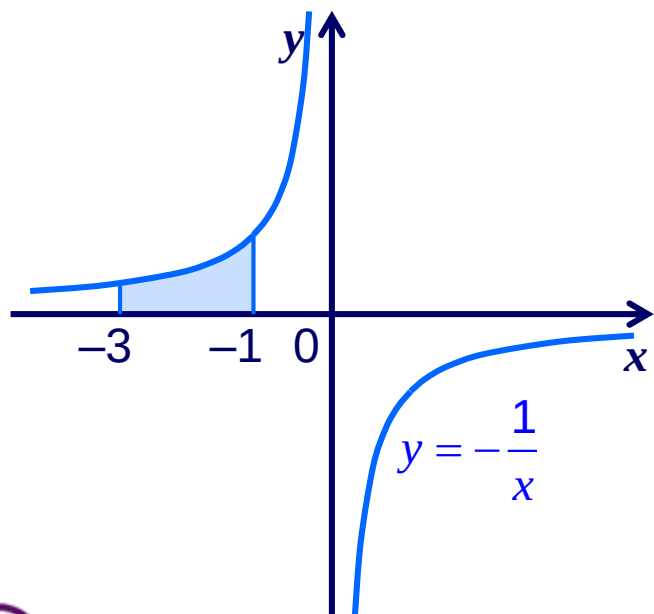
Find $\int 4x \left(x^2 - \frac{5}{x^2} \right) dx$

$$\begin{aligned} \int 4x \left(x^2 - \frac{5}{x^2} \right) dx &= \int 4x^3 - \frac{20}{x} dx \\ &= x^4 - 20 \ln|x| + c \end{aligned}$$

Definite integrals involving $\frac{1}{x}$

It is particularly important to remember the modulus sign when evaluating definite integrals of functions involving $\frac{1}{x}$.

Find the area under the curve $y = -\frac{1}{x}$ between $x = -3$, $x = -1$ and the x -axis, writing your answer in the form $\ln a$.



The area is given by $\int_{-3}^{-1} -\frac{1}{x}$.

$$\int_{-3}^{-1} -\frac{1}{x} = \left[-\ln|x| \right]_{-3}^{-1}$$

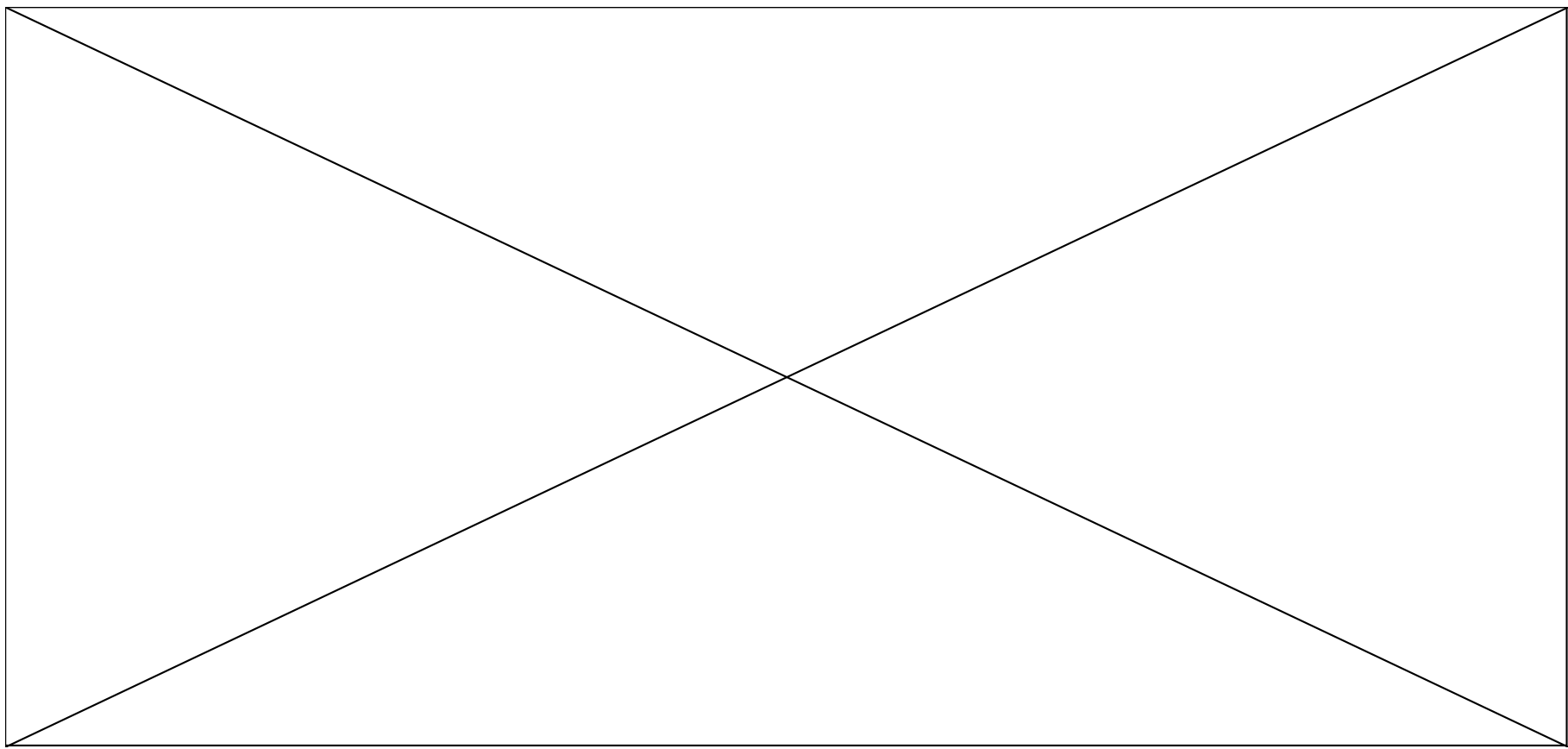
$$= -\ln|-1| - -\ln|-3|$$

$$= -\ln 1 + \ln 3$$

$$= \ln 3 \text{ units squared}$$

Remember
that $\ln 1 = 0$

We should note that definite integrals of the form $\int_a^b \frac{1}{x}$ can only be evaluated if $x = 0$ does not lie in the interval $[a, b]$.



Integrals of standard functions

By reversing the process of differentiation we can derive the integrals of some standard functions.

$$\int x^n dx = \frac{x^{n+1}}{n+1} + c \quad (n \neq -1)$$

$$\int \sin x dx = -\cos x + c$$

$$\int \frac{1}{x} dx = \ln|x| + c$$

$$\int \cos x dx = \sin x + c$$

$$\int e^x dx = e^x + c$$

$$\int \sec^2 x dx = \tan x + c$$

These integrals should be memorized.



Also, if any function is multiplied by a constant k then its integral will also be multiplied by the constant k .

Find $\int (4 \sin x + 7e^x) dx$

$$\begin{aligned}\int (4 \sin x + 7e^x) dx &= \int 4 \sin x dx + \int 7e^x dx \\ &= 4 \int \sin x dx + 7 \int e^x dx \\ &= 4(-\cos x) + 7e^x \\ &= 7e^x - 4\cos x\end{aligned}$$

In practice most of these steps can be left out.



- Integrals of standard functions
- **Reversing the chain rule**
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A very helpful technique is to recognize that a function that we are trying to integrate is of a form given by the differentiation of a composite function. This is sometimes called **integration by recognition**.

Let $y = (f(x))^{n+1}$

By the chain rule: $\frac{dy}{dx} = (n+1)(f(x))^n f'(x)$

So $\int (n+1)(f(x))^n f'(x) dx = (f(x))^{n+1} + c$

It follows that for $n \neq -1$

$$\int (f(x))^n f'(x) dx = \frac{1}{(n+1)} (f(x))^{n+1} + c$$

Reversing the chain rule

If the integral is multiplied by a constant k :

$$\int k(f(x))^n f'(x) dx = \frac{k}{(n+1)} (f(x))^{n+1} + c$$

Don't try to learn this formula, just try to recognize that the function you are integrating is of the form $k(f(x))^n f'(x)$ and compare it to the derivative of $(f(x))^{n+1}$.

Suppose we want to integrate $(2x + 7)^5$ with respect to x .

Consider the derivative of $y = (2x + 7)^6$.

Using the chain rule: $\frac{dy}{dx} = 6(2x + 7)^5 \times 2 = 12(2x + 7)^5$

So $\int 12(2x + 7)^5 dx = (2x + 7)^6 + c$

$$\therefore \int (2x + 7)^5 dx = \frac{1}{12} (2x + 7)^6 + c$$

In general, you can integrate any linear function raised to a power using the formula:

$$\int (ax + b)^n dx = \frac{1}{a(n+1)} (ax + b)^{n+1} + c$$

With practice, integrals of this type can be written down directly. For example:

$$\int (x - 5)^8 dx = \frac{1}{9} (x - 5)^9 + c$$

$$\int (4x + 7)^2 dx = \frac{1}{12} (4x + 7)^3 + c$$

$$\int 5(3 - 2x)^9 dx = \frac{5}{-20} (3 - 2x)^{10} + c = \frac{1}{-4} (3 - 2x)^{10} + c$$

Reversing the chain rule

Let's look at some more integrals of functions of the form $k(f(x))^n f'(x)$.

Integrate $y = x(3x^2 + 4)^3$ with respect to x .

Notice that the derivative of $3x^2 + 4$ is $6x$.

Now consider the derivative of $y = (3x^2 + 4)^4$.

Using the chain rule: $\frac{dy}{dx} = 4(3x^2 + 4)^3 \times 6x = 24x(3x^2 + 4)^3$

So
$$\int 24x(3x^2 + 4)^3 dx = (3x^2 + 4)^4 + c$$

$$\therefore \int x(3x^2 + 4)^3 dx = \frac{1}{24} (3x^2 + 4)^4 + c$$

Reversing the chain rule

Find $\int 7x^2(2x^3 - 9)^2 dx$

Notice that the derivative of $2x^3 - 9$ is $6x^2$.

Now consider the derivative of $y = (2x^3 - 9)^3$.

Using the chain rule: $\frac{dy}{dx} = 3(2x^3 - 9)^2 \times 6x^2 = 18x^2(2x^3 - 9)^2$

So
$$\int 18x^2(2x^3 - 9)^2 dx = (2x^3 - 9)^3 + c$$

$$\Rightarrow \int x^2(2x^3 - 9)^2 dx = \frac{1}{18}(2x^3 - 9)^3 + c$$

$$\therefore \int 7x^2(2x^3 - 9)^2 dx = \frac{7}{18}(2x^3 - 9)^3 + c$$

Reversing the chain rule

Find $\int \frac{x^2}{\sqrt{x^3 - 1}} dx$.

x^2 is the derivative of $(x^3 - 1)$.

Start by writing $\frac{x^2}{\sqrt{x^3 - 1}}$ as $x^2(x^3 - 1)^{-\frac{1}{2}}$.

$-\frac{1}{2}$ plus 1 is $\frac{1}{2}$.

Now consider the derivative of $y = (x^3 - 1)^{\frac{1}{2}}$.

Using the chain rule: $\frac{dy}{dx} = \frac{1}{2}(x^3 - 1)^{-\frac{1}{2}} \times 3x^2 = \frac{3}{2}x^2(x^3 - 1)^{-\frac{1}{2}}$

So $\int \frac{3}{2}x^2(x^3 - 1)^{-\frac{1}{2}} dx = (x^3 - 1)^{\frac{1}{2}} + c$

$$\Rightarrow \int x^2(x^3 - 1)^{-\frac{1}{2}} dx = \frac{2}{3}(x^3 - 1)^{\frac{1}{2}} + c$$

$$\therefore \int \frac{x^2}{\sqrt{x^3 - 1}} dx = \frac{2\sqrt{x^3 - 1}}{3} + c$$

When we applied the chain rule to functions of the form $e^{f(x)}$ we obtained the following generalization:

$$\text{If } y = e^{f(x)} \text{ then } \frac{dy}{dx} = f'(x)e^{f(x)}$$

We can reverse this to integrate functions of the form $kf'(x)e^{f(x)}$.
For example:

$$\int 5e^{5x-3} dx = e^{5x-3} + c$$

A numerical adjustment is usually necessary.

$$\int e^{5x-3} dx = \frac{1}{5}e^{5x-3} + c$$

$$\int 2e^{5x-3} dx = \frac{2}{5}e^{5x-3} + c$$

Reversing the chain rule for exponential functions

In general,

$$\int e^{ax+b} dx = \frac{1}{a} e^{ax+b} + c$$

Find $\int \frac{6}{e^{3x}} dx$.

$$\int \frac{6}{e^{3x}} dx = \int 6e^{-3x} dx$$

$$= \frac{6}{-3} e^{-3x} + c$$

$$= -2e^{-3x} + c$$

$$= -\frac{2}{e^{3x}} + c$$

With practice, this method can be extended to cases where the exponent is not linear. For example:

Find $\int xe^{2x^2} dx$.

Notice that the derivative of $2x^2$ is $4x$ and so the function we are integrating is of the form $kf'(x)e^{f(x)}$.

$$\begin{aligned}\int xe^{2x^2} dx &= \frac{1}{4} \int 4xe^{2x^2} dx \\ &= \frac{1}{4} e^{2x^2} + c\end{aligned}$$

When we applied the chain rule to functions of the form $\ln f(x)$ we obtained the following generalization:

$$\text{If } y = \ln f(x) \text{ then } \frac{dy}{dx} = \frac{f'(x)}{f(x)}$$

We can reverse this to integrate functions of the form $\frac{kf'(x)}{f(x)}$
For example:

$$\begin{aligned} \int \frac{1}{5x+4} dx &= \frac{1}{5} \int \frac{5}{5x+4} dx \\ &= \frac{1}{5} \ln|5x+4| + c \end{aligned}$$

Remove a factor of $\frac{1}{5}$ to write the function in the form $\frac{f'(x)}{f(x)}$.

In general,

$$\int \frac{1}{ax+b} dx = \frac{1}{a} \ln|ax+b| + c$$

Find $\int \frac{5}{2-6x} dx$.

This is now of the form $\frac{f'(x)}{f(x)}$.

$$\begin{aligned}\int \frac{5}{2-6x} dx &= -\frac{5}{6} \int \frac{-6}{2-6x} dx \\ &= -\frac{5}{6} \ln|2-6x| + c\end{aligned}$$

Evaluate $\int_{-1}^2 \frac{4}{2x-7} dx$, writing your answer in the form $a \ln b$.

First of all, note that the graph of $y = \frac{4}{2x-7}$ has a discontinuity when $2x - 7 = 0$, that is when $x = 3.5$.

This is outside the interval $[-1, 2]$ and so the integral is valid.

Reversing the chain rule for logarithmic functions

$$\int_{-1}^2 \frac{4}{2x-7} dx = 2 \int_{-1}^2 \frac{2}{2x-7} dx$$

This is now of the form $\frac{f'(x)}{f(x)}$.

$$= 2 \left[\ln|2x-7| \right]_{-1}^2$$

$$= 2(\ln|4-7| - \ln|-2-7|)$$

$$= 2(\ln 3 - \ln 9)$$

This can be written in the required form by using the rule that $\ln a - \ln b = \ln \frac{a}{b}$.

$$\begin{aligned} \int_{-1}^2 \frac{4}{2x-7} dx &= 2 \ln \frac{3}{9} \\ &= 2 \ln \frac{1}{3} \end{aligned}$$

Reversing the chain rule for logarithmic functions

Find $\int \frac{x^2}{7+2x^3} dx$.

This is now of
the form $\frac{f'(x)}{f(x)}$.

$$\begin{aligned}\int \frac{x^2}{7+2x^3} dx &= \frac{1}{6} \int \frac{6x^2}{7+2x^3} dx \\ &= \frac{1}{6} \ln|7+2x^3| + c\end{aligned}$$

Find $\int \frac{4e^{2x}}{e^{2x}+1} dx$.

This is now of
the form $\frac{f'(x)}{f(x)}$.

$$\begin{aligned}\int \frac{4e^{2x}}{e^{2x}+1} dx &= 2 \int \frac{2e^{2x}}{e^{2x}+1} dx \\ &= 2 \ln|e^{2x}+1| + c\end{aligned}$$

The integral of $\tan x$

We can find the integral of $\tan x$ by writing it as $\frac{\sin x}{\cos x}$ and recognizing that this fraction is of the form $\frac{-f'(x)}{f(x)}$.

$$\begin{aligned}\int \tan x \, dx &= \int \frac{\sin x}{\cos x} \, dx \\ &= -\ln|\cos x| + c\end{aligned}$$

It is 'tidier' to rewrite this without a minus sign at the front, using the fact that $-\ln a = \ln a^{-1}$:

$$\begin{aligned}-\ln|\cos x| + c &= \ln|\cos x|^{-1} + c \\ &= \ln \frac{1}{|\cos x|} + c \\ &= \ln|\sec x| + c\end{aligned}$$

$$\int \tan x \, dx = \ln|\sec x| + c$$

When we applied the chain rule to functions of the form $\sin f(x)$ and $\cos f(x)$ we obtained the following generalizations:

$$\text{If } y = \sin f(x) \text{ then } \frac{dy}{dx} = f'(x) \cos f(x)$$

$$\text{If } y = \cos f(x) \text{ then } \frac{dy}{dx} = -f'(x) \sin f(x)$$

We can reverse these to integrate functions of the form $f'(x) \cos f(x)$ and $f'(x) \sin f(x)$. For example:

$$\int 3 \cos 3x \, dx = \sin 3x + c$$

$$\int -2x \sin x^2 \, dx = \cos x^2 + c$$

As with other examples a numerical adjustment is often necessary.

$$\begin{aligned}\int \sin(5x-2)dx &= -\frac{1}{5} \int -5 \sin(5x-2)dx \\ &= -\frac{1}{5} \cos(5x-2)dx\end{aligned}$$

This is now of the form $-f'(x) \sin f(x)$.

In general, when dealing with the cos and sin of linear functions:

$$\int \cos(ax+b)dx = \frac{1}{a} \sin(ax+b) + c$$

$$\int \sin(ax+b)dx = -\frac{1}{a} \cos(ax+b) + c$$

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With practice, the technique of **integration by recognition** can save a lot of time.

However, when it is too difficult to use integration by recognition we can use a more formal method of reversing the chain rule called **integration by substitution**.

To see how this method works consider the integral $\int (5x + 2)^3 dx$.

Let $u = 5x + 2$ so that $\int (5x + 2)^3 dx = \int u^3 dx$

The problem now is that we can't integrate a function in u with respect to x . We therefore need to write dx in terms of du .

$$u = 5x + 2 \Rightarrow \frac{du}{dx} = 5$$



When we used the chain rule for differentiation we saw that we can treat $\frac{du}{dx}$ informally as a fraction, so:

$$\frac{du}{dx} = 5 \Rightarrow du = 5dx \Rightarrow dx = \frac{1}{5} du$$

So if $u = 5x + 2$ and $dx = \frac{1}{5} du$:

$$\begin{aligned} \int (5x + 2)^3 dx &= \int u^3 \frac{1}{5} du \\ &= \frac{1}{20} u^4 + c \end{aligned}$$

Now change the variable back to x :

$$\int (5x + 2)^3 dx = \frac{1}{20} (5x + 2)^4 + c$$

Use a suitable substitution to find $\int x(2x^2 - 5)^5 dx$.

$$\begin{aligned}\text{Let } u &= 2x^2 - 5 & \frac{du}{dx} &= 4x \\ & & \frac{dx}{du} &= \frac{1}{4x} \\ & & dx &= \frac{1}{4x} du\end{aligned}$$

Substituting u and dx into the original problem gives:

$$\begin{aligned}\int x(2x^2 - 5)^5 dx &= \int x u^5 \frac{1}{4x} du \quad \leftarrow \text{Notice that the } x\text{'s cancel out.} \\ &= \frac{1}{4} \int u^5 du \\ &= \frac{1}{24} u^6 + c\end{aligned}$$

Now we need to change the variable back to x :

$$\int x(2x^2 - 5)^5 dx = \frac{1}{24} (2x^2 - 5)^6 + c$$

This integral could also have been found directly by recognition.

However, there are functions that can be integrated by use of a suitable substitution but not by recognition. For instance:

Use the substitution $u = 1 - 2x$ to find $\int x(1 - 2x)^4 dx$.

If $u = 1 - 2x$ then $\frac{du}{dx} = -2$

$$\frac{dx}{du} = -\frac{1}{2}$$

$$dx = -\frac{1}{2} du$$

Also if $u = 1 - 2x$ then

$$x = \frac{1}{2}(1 - u) \quad \leftarrow$$

We also have to substitute the x so that the whole integrand is in terms of u .

Substituting these into the original problem gives:

$$\begin{aligned} \int x(1 - 2x)^4 dx &= \int \frac{1}{2}(1 - u)u^4 \left(-\frac{1}{2}\right) du \\ &= -\frac{1}{4} \int (1 - u)u^4 du \\ &= -\frac{1}{4} \int (u^4 - u^5) du \end{aligned}$$



$$\begin{aligned} -\frac{1}{4} \int (u^4 - u^5) du &= -\frac{1}{4} \left(\frac{u^5}{5} - \frac{u^6}{6} \right) + c \\ &= -\frac{1}{4} \left(\frac{6u^5 - 5u^6}{30} \right) + c \\ &= -\frac{1}{120} u^5 (6 - 5u) + c \end{aligned}$$

Changing the variable back to x gives:

$$\begin{aligned} \int x(1-2x)^4 dx &= -\frac{1}{120} (1-2x)^5 (6 - 5(1-2x)) + c \\ &= -\frac{1}{120} (1-2x)^5 (10x + 1) + c \\ &= \frac{1}{120} (2x - 1)^5 (10x + 1) + c \end{aligned}$$



Definite integration by substitution

When a definite integral is found by substitution it is easiest to rewrite the limits of integration in terms of the substituted variable.

Use the substitution $u = \sqrt{8-x}$ to find the area under the curve $y = \frac{2x}{\sqrt{8-x}}$ between $x = 4$ and $x = 7$.

If $u = (8-x)^{\frac{1}{2}}$ then

$$\frac{du}{dx} = \frac{1}{2}(8-x)^{-\frac{1}{2}}(-1) \leftarrow \text{Using the chain rule for differentiation.}$$

$$= -\frac{1}{2(8-x)^{\frac{1}{2}}}$$

$$= -\frac{1}{2u}$$

So $\frac{dx}{du} = -2u \Rightarrow dx = -2u du$

Definite integration by substitution

Now we need to find x in terms of u .

If $u = (8 - x)^{\frac{1}{2}}$ then $u^2 = 8 - x$

$$x = 8 - u^2$$

Rewrite the limits in terms of u :

$$\text{when } x = 3, u = \sqrt{8 - 4} = 2$$

$$\text{when } x = 1, u = \sqrt{8 - 7} = 1$$

The area is given by $\int_4^7 \frac{2x}{\sqrt{8-x}} dx$. Rewrite this in terms of u :

$$\begin{aligned} \int_4^7 \frac{2x}{\sqrt{8-x}} dx &= \int_1^2 \frac{2(8-u^2)}{u} (-2u) du \\ &= \int_1^2 -4(8-u^2) du \\ &= 4 \int_1^2 (u^2 - 8) du \end{aligned}$$

Definite integration by substitution

$$\begin{aligned}4 \int_1^2 (u^2 - 8) du &= 4 \left[\frac{u^3}{3} - 8u \right]_1^2 \\&= 4 \left(\frac{8}{3} - 16 - \frac{1}{3} + 8 \right) \\&= 4 \left(\frac{7}{3} - 8 \right) \\&= -22\frac{2}{3}\end{aligned}$$

Therefore, the required area is $22\frac{2}{3}$ units squared.



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Suppose we wish to integrate the product of two functions, such as $x \sin x$, where one of the functions is not related to the derivative of the other.

An expression such as this can be integrated using the method of **integration by parts**.

When we *differentiate* the product of two functions u and v we use the product rule:

$$\frac{d}{dx}(uv) = u \frac{dv}{dx} + v \frac{du}{dx}$$

where u and v are functions of x .

Integrating throughout with respect to x gives:

$$uv = \int u \frac{dv}{dx} dx + \int v \frac{du}{dx} dx$$

This can be rearranged to give:

$$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$$

To integrate a product using this formula we let one part equal u and the other equal $\frac{dv}{dx}$.

We find $\frac{du}{dx}$ by differentiating the part we called u .

We find v by integrating the part we called $\frac{dv}{dx}$.

It is important to choose u and $\frac{dv}{dx}$ so that

$\int v \frac{du}{dx} dx$ is easier to integrate than $\int u \frac{dv}{dx} dx$.



Integration by parts

So, to integrate $x \sin x$ with respect to x :

Let $u = x$ and $\frac{dv}{dx} = \sin x$

So $\frac{du}{dx} = 1$ and $v = -\cos x$

We don't need the "+ c" here.

Now, using the formula $\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$:

$$\int x \sin x dx = -x \cos x - \int -\cos x dx$$

$$= -x \cos x - -\sin x + c$$

$$= \sin x - x \cos x + c$$

Find $\int xe^{2x} dx$.

Let $u = x$ and $\frac{dv}{dx} = e^{2x}$

So $\frac{du}{dx} = 1$ and $v = \frac{e^{2x}}{2}$

Now, apply the formula $\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$:

$$\begin{aligned}\int xe^{2x} dx &= x \frac{e^{2x}}{2} - \int \frac{e^{2x}}{2} dx \\ &= \frac{xe^{2x}}{2} - \frac{e^{2x}}{4} + c \\ &= \frac{e^{2x}}{4} (2x - 1) + c\end{aligned}$$

Find $\int 8x \ln x \, dx$.

We don't know the integral of $\ln x$ so:

let $u = \ln x$ and $\frac{dv}{dx} = 8x$

so $\frac{du}{dx} = \frac{1}{x}$ and $v = 4x^2$

Now, using the formula, $\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$:

$$\int 8x \ln x \, dx = 4x^2 \ln x - \int 4x^2 \times \frac{1}{x} dx$$

$$= 4x^2 \ln x - \int 4x \, dx$$

$$= 4x^2 \ln x - 2x^2 + c$$

$$= 2x^2(2\ln x - 1) + c$$

The integral of $\ln x$

We can also use integration by parts to find the integral of $\ln x$.

We do this by writing $\ln x$ as $(1 \times \ln x)$.

Let $u = \ln x$ and $\frac{dv}{dx} = 1$

So $\frac{du}{dx} = \frac{1}{x}$ and $v = x$

Now we can integrate by parts:

$$\begin{aligned}\int 1 \times \ln x \, dx &= x \ln x - \int x \frac{1}{x} \, dx \\ &= x \ln x - \int 1 \, dx \\ &= x \ln x - x + c\end{aligned}$$

$$\int \ln x \, dx = x \ln x - x + c$$

Integration by parts

To evaluate a definite integral using integration by parts we use:

$$\int_a^b u \frac{dv}{dx} dx = [uv]_a^b - \int_a^b v \frac{du}{dx} dx$$

Evaluate $\int_0^{\frac{\pi}{2}} 2x \cos x dx$

Let $u = 2x$ and $\frac{dv}{dx} = \cos x$

So $\frac{du}{dx} = 2$ and $v = \sin x$

$$\int_0^{\frac{\pi}{2}} 2x \cos x dx = [2x \sin x]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} 2 \sin x dx$$

$$\sin \pi/2 = 1 \quad \quad \quad = (\pi \sin \frac{\pi}{2} - 0) - [-2 \cos x]_0^{\frac{\pi}{2}}$$

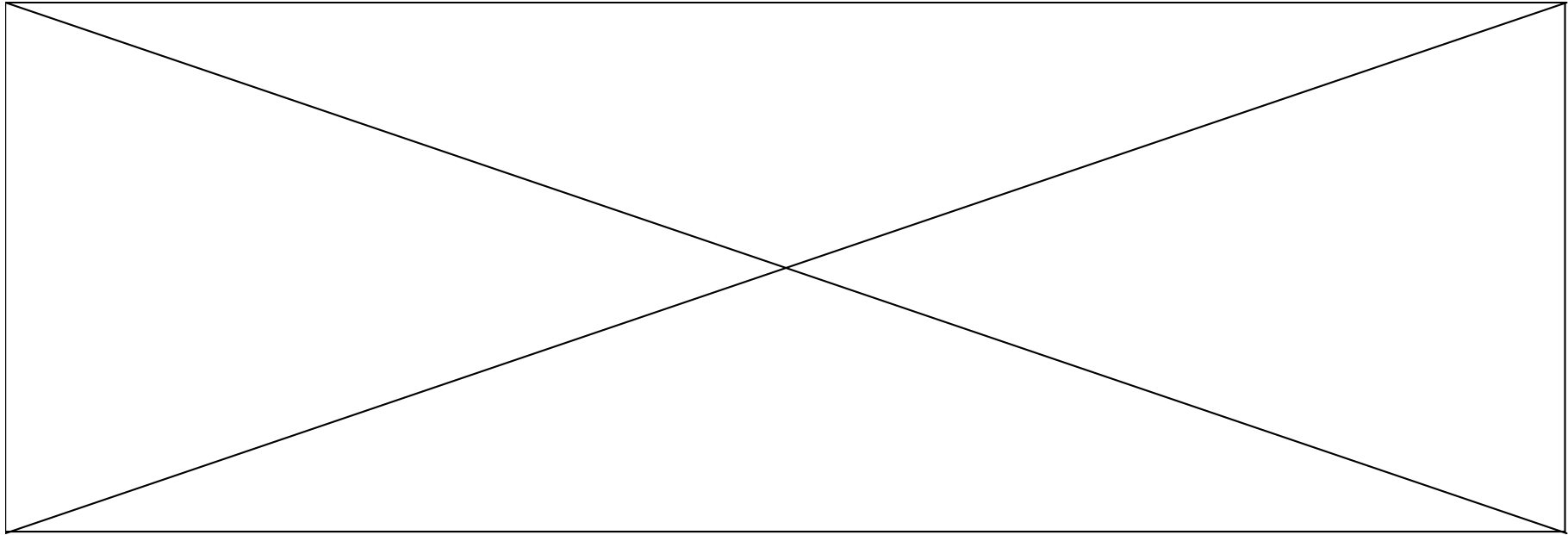
$$\cos \pi/2 = 0 \quad \quad \quad = \pi - (-2 \cos \frac{\pi}{2} + 2 \cos 0)$$

$$\cos 0 = 1 \quad \quad \quad = \pi - 2$$

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- Examination-style question



Consider the area bounded by the curve $y = f(x)$, the x -axis and $x = a$ and $x = b$.

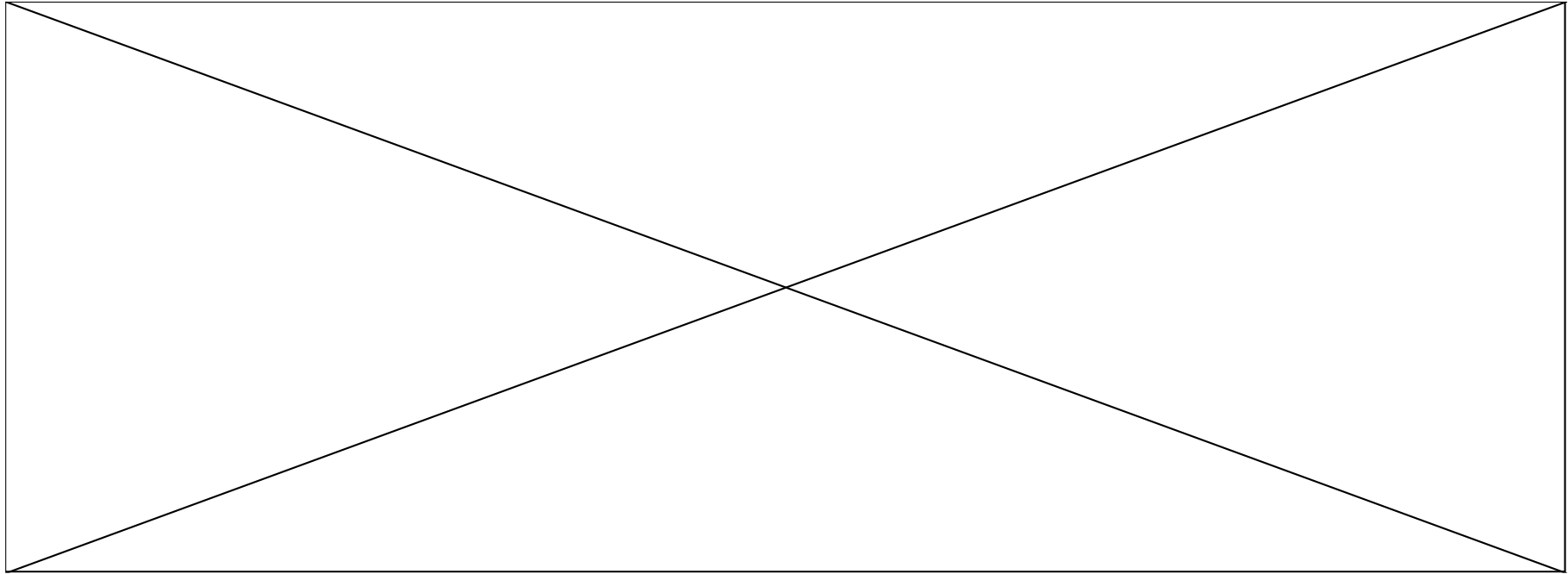


If this area is rotated 360° about the x -axis a three-dimensional shape called a **solid of revolution** is formed.

The volume of this solid is called its **volume of revolution**.



We can calculate the volume of revolution by dividing the volume of revolution into thin slices of width δx .



The volume of each slice is approximately cylindrical, of radius y and height δx , and is therefore approximately equal to

$$\pi y^2 \delta x$$



The total volume of the solid is given by the sum of the volume of the slices.

$$V \approx \sum_{x=a}^{x=b} \pi y^2 \delta x$$

The smaller δx is, the closer this approximate area is to the actual area.

We can find the *actual* area by considering the limit of this sum as δx tends to 0.

$$V = \lim_{\delta x \rightarrow 0} \sum_{x=a}^{x=b} \pi y^2 \delta x$$

This limit is represented by the following integral:

$$V = \int_a^b \pi y^2 dx$$



So in general, the volume of revolution V of the solid generated by rotating the curve $y = f(x)$ between $x = a$ and $x = b$ about the x -axis is:

$$V = \int_a^b \pi y^2 dx$$

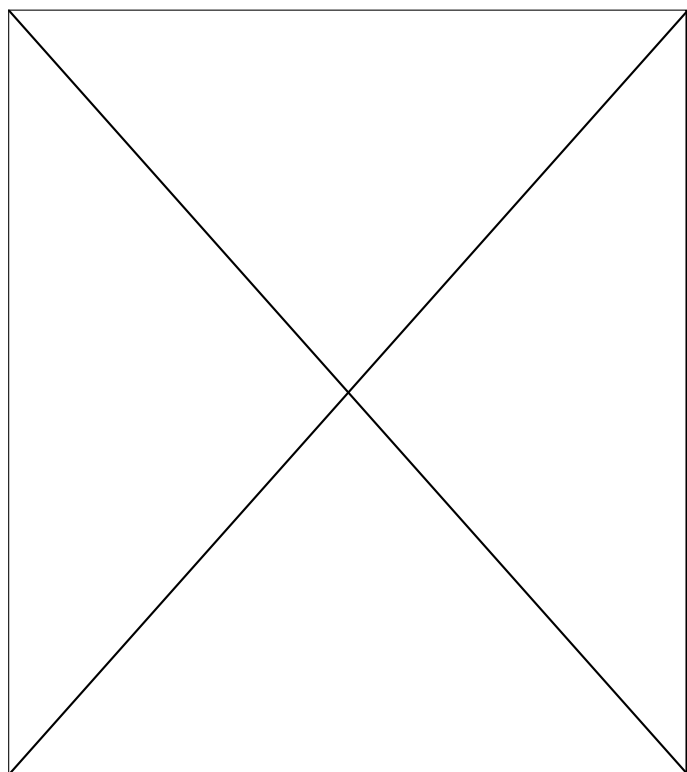
Similarly, the volume of revolution V of the solid generated by rotating the curve $x = f(y)$ between $y = a$ and $y = b$ about the y -axis is:

$$V = \int_a^b \pi x^2 dy$$

Volumes of revolution are usually given as multiples of π .

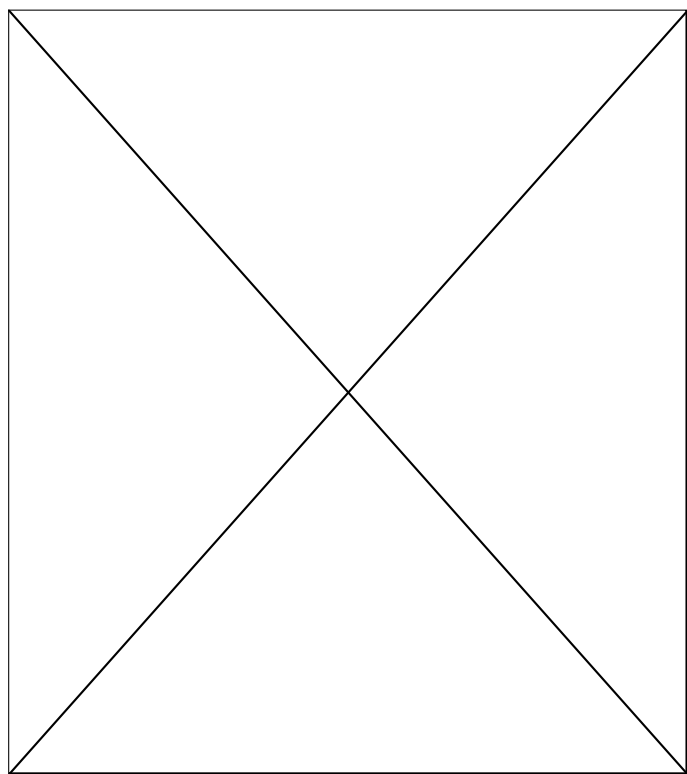


Find the volume of the solid formed by rotating the area between the curve $y = x(2 - x)$, the x -axis, $x = 0$, and $x = 2$ 360° about the x -axis.



$$\begin{aligned} V &= \int_0^2 \pi y^2 dx \\ &= \pi \int_0^2 (2x - x^2)^2 dx \\ &= \pi \int_0^2 (4x^2 - 4x^3 + x^4) dx \\ &= \pi \left[\frac{4x^3}{3} - \frac{4x^4}{4} + \frac{x^5}{5} \right]_0^2 \\ &= \pi \left(\frac{32}{3} - 16 + \frac{32}{5} \right) = \frac{16\pi}{15} \text{ cubic units} \end{aligned}$$

Find the volume of the solid formed by rotating the area between the curve $y = \frac{1}{x}$, the y -axis, $y = 1$, and $y = 2$ 360° about the y -axis.



$$V = \int_1^2 \pi x^2 dy$$

$$= \pi \int_1^2 \left(\frac{1}{y} \right)^2 dy$$

Rearranging
 $y = \frac{1}{x}$ gives $x = \frac{1}{y}$.

$$= \pi \int_1^2 y^{-2} dy$$

$$= \pi \left[-\frac{1}{x} \right]_1^2$$

$$= \pi \left(-\frac{1}{2} + 1 \right) = \frac{\pi}{2} \text{ cubic units}$$

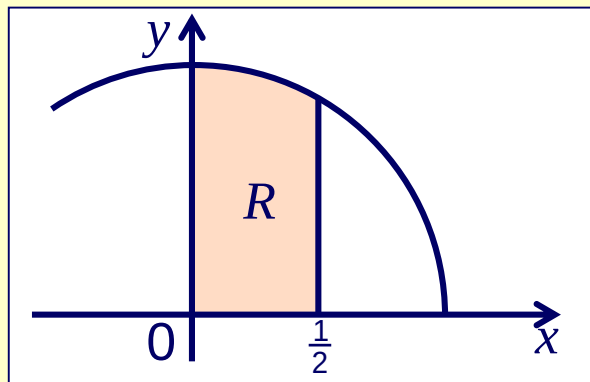
- Integrals of standard functions
- Reversing the chain rule
- Integration by substitution
- Integration by parts
- Volumes of revolution
- **Examination-style question**



Examination-style question

a) Show that $\cos^2 x = \frac{1}{2}(1 + \cos 2x)$

b) The diagram below shows the finite region R which is bounded by the curve $y = \sqrt{1-x^2}$ the x -axis, the y -axis and the line $x = \frac{1}{2}$.



Using the substitution $x = \sin u$, or otherwise, find the area of the region R to 3 significant figures.

c) Find the volume of the solid formed when R is rotated through 360° about the x -axis to 3 significant figures.

Examination-style question

a) $\text{RHS} = \frac{1}{2}(1 + \cos 2x)$

$$= \frac{1}{2}(1 + (2\cos^2 x - 1))$$

$$= \frac{1}{2}(2\cos^2 x)$$

$$= \cos^2 x = \text{LHS}$$

Using the double angle
identity $\cos 2A \equiv 2\cos^2 A - 1$.

b) The area of R is given by

$$\int_0^{\frac{1}{2}} \sqrt{1-x^2} dx$$

Using the substitution $x = \sin u$ we have:

$$\frac{dx}{du} = \cos u \Rightarrow dx = \cos u du$$

Find u when x is 0 and $\frac{1}{2}$:

when $x = 0$, $u = 0$

and when $x = \frac{1}{2}$, $u = \frac{\pi}{6}$

Examination-style question

Now make the substitution:

$$\begin{aligned}\int_0^{\frac{1}{2}} \sqrt{1-x^2} \, dx &= \int_0^{\frac{\pi}{6}} \sqrt{1-\sin^2 u} \cos u \, du \\ &= \int_0^{\frac{\pi}{6}} \cos^2 u \, du \quad \leftarrow \text{Using the identity } \cos^2 A = 1 - \sin^2 A. \\ &= \int_0^{\frac{\pi}{6}} \frac{1}{2} (1 + \cos 2u) \, du \quad \leftarrow \text{Using the identity } \cos^2 A = \frac{1}{2} (1 + \cos 2A). \\ &= \frac{1}{2} \left[u + \frac{1}{2} \sin 2u \right]_0^{\frac{\pi}{6}} \\ &= \frac{\pi}{12} + \frac{\sqrt{3}}{8}\end{aligned}$$

Therefore, the area of region R is **0.478 units squared** (to 3 s.f.).



c) The required volume of revolution is given by

$$\begin{aligned}\int_0^{\frac{1}{2}} \pi y^2 dx &= \pi \int_0^{\frac{1}{2}} (1-x^2) dx \\&= \pi \left[x - \frac{x^3}{3} \right]_0^{\frac{1}{2}} \\&= \pi \left(\frac{1}{2} - \frac{\frac{1}{8}}{3} \right) \\&= \frac{11\pi}{24}\end{aligned}$$

Therefore, the volume formed when R is rotated about the x -axis is **1.44 units cubed** (to 3 s.f.).

